

Exercise 3.7

$$a) \frac{8.765 \times \sqrt[3]{27.03}}{24.51} = x$$

$$\lg x = \lg 8.765 + \lg 27.03^{\frac{1}{3}} - \lg 24.51$$

$$= 0.9427 + 1.2700^{\frac{1}{3}} - 1.2451$$

$$= 0.9427 + \frac{1}{3} \times 1.4319 - 1.3894$$

$$= 0.9427 + 0.4768 - 1.3894$$

$$\lg x = 0.0301 \text{ (A. log)}$$

$$x = 3.01 \times 10^{-2} = 1.071 \times 10^0$$

$$= 1.071 \text{ (A. log)}$$

$$= 1.071 \approx 1.071 \text{ (A. log)}$$

$$a) \frac{8.765 \times \sqrt[3]{27.03}}{24.51}$$

$$x = \frac{8.765 \times \sqrt[3]{27.03}}{24.51}$$

$$\lg x = \lg 8.765 + \lg 27.03^{\frac{1}{3}} - \lg 24.51$$

$$= \lg 0.9427 + \frac{1}{3} \times 1.4319 - 1.3894$$

$$= \lg 0.9427 + 0.4773 - 1.3894$$

$$= 1.4200 - 1.3894$$

$$= 0.0306 \text{ (A. log)}$$

$$= 1.073 \times 10^0$$

$$x = 1.073$$

$$\begin{array}{r} 0.9427 \\ 0.4768 \\ \hline 1.4195 \\ 1.3894 \\ \hline 0.0301 \end{array}$$

$$\begin{array}{r} 0.4768 \\ 3 \overline{) 1.4314} \\ \underline{12} \\ 23 \\ \underline{21} \\ 20 \\ \underline{18} \\ 29 \\ \underline{26} \\ 3 \end{array}$$

$$\begin{array}{r} 0. \\ 3 \overline{) 1.4314} \\ \underline{0} \end{array}$$

$$\begin{array}{r} 0.4773 \\ 3 \overline{) 1.4319} \\ \underline{12} \\ 23 \\ \underline{21} \\ 21 \\ \underline{21} \\ 09 \\ \underline{9} \end{array}$$

$$\begin{array}{r} (-1) + 1 \\ -1 + 1 \end{array}$$

$$b) \frac{\sqrt{9.18 \times 8.02^2}}{9.83} = x$$

$$\lg x = \lg \left(\frac{9.18^{\frac{1}{2}} \times 8.02^2}{9.83} \right)$$

$$\lg x = \frac{1}{2} \times \lg 9.18 + 2 \times \lg 8.02 - \lg 9.83$$

$$\lg x = \frac{1}{2} \times 0.9628 + 2 \times 0.9042 - 0.9926$$

$$\lg x = 0.4814 + 1.8084 - 0.9926$$

$$= 2.2898 - 0.9926$$

$$x = 1.2972 \text{ (A. log)}$$

$$x = 1.982 \times 10^1$$

$$= 19.82$$

$$c) \frac{\sqrt{0.0945 \times 4.821^2}}{48.15} = x$$

$$\lg x = \lg \left(\frac{0.0945^{\frac{1}{2}} \times 4.821^2}{48.15} \right)$$

$$\lg x = \frac{1}{2} \times \lg 0.0945 + 2 \times \lg 4.821 - \lg 48.15$$

$$= \frac{1}{2} \times 2.9754 + 2 \times 0.6831 - 1.6825$$

$$= 1.4877 + 1.3662 - 1.6825$$

$$= 0.8499 - 1.6825$$

$$=$$

$$\begin{array}{r} 1.4877 \\ 2 \overline{) 2.9754} \\ \underline{2} \\ 09 \\ 8 \\ 17 \\ 16 \\ 15 \\ 14 \\ 14 \end{array}$$

$$\begin{array}{r} 1.4877 \\ + 1.3622 \\ \hline 0.8499 \\ - 1.6825 \\ \hline 1.674 \end{array}$$

$$\begin{array}{r} 0.8499 \\ - 1.6825 \\ \hline 1.674 \end{array}$$

$$d) \frac{3 \times 0.752^2}{\sqrt{17.96}} = x$$

$$\lg x = \lg \left(3000 + 0.752^2 - 17.96^{\frac{1}{2}} \right)$$

$$\lg x = \lg 3.000 + 2 \times \lg 0.752 - \frac{1}{2} \times 17.96$$

$$= \lg 0.4771 + 2 \times \bar{1}.8762 - \frac{1}{2} \times 1.2544$$

$$= 0.4771 + \bar{1}.7524 - 0.6272$$

$$= 0.2295 - 0.6272$$

$$= \bar{1}.5423 \text{ (A. log)}$$

$$= 3.486 \times 10^{-1} = 0.3486$$

$$e) \frac{6.591 \times \sqrt[3]{0.0782}}{0.9821^2} = x$$

$$\lg x = \lg \left(6.591 \times 0.0782^{\frac{1}{3}} \div 0.9821^2 \right)$$

$$= \lg 0.8190 + \frac{1}{3} \times \bar{2}.8932 - 2 \times \bar{1}.9921$$

$$= 0.8190 + \bar{2}.2977 - \bar{1}.9842$$

$$f) \frac{3.251 \times \sqrt[3]{0.0234}}{0.8915} = x$$

$$\lg x = \lg \left(3.251 \times 0.0234^{\frac{1}{3}} \div 0.8915 \right)$$

$$= \lg 3.251 + \lg 0.5120 + \lg \frac{1}{3} \times \lg 2.3692 - \lg 0.8915$$

$$= \lg 0.5120 + \bar{1}.4564 - \bar{1}.9501$$

$$= 0.0684 - \bar{1}.9501$$

$$= 0.1183 \text{ (A. log)}$$

$$= 1.313 \times 10^0$$

$$= 1.313$$